

QUESTIONS 10 / 10 completed

Binary Tree-Inorder

BINARY TREE-INORDER REVERSAL

Write a C program to create a Binary Tree and perform Recursive Inorder Traversal. Consider an organizational hierarchy management system where employee records are stored in a binary tree structure. The program should create the tree dynamically and

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 #include <malloc.h>
3
4 struct node
5 {
6     struct node* left;
7     int data;
8     struct node* right;
9 }*root = NULL, *temp;
10
11 struct node* create(struct node *root, int ele)
12 {
13     temp = (struct node*)malloc(sizeof(struct node));
14     temp->data = ele;
15     temp->left = NULL;
16     temp->right = NULL;
17
18     if (root == NULL)
19     {
20         root = temp;
21     }
22     return root;
```

OUTPUT

```
inorder: 40 50 55 60
preorder: 50 40 60 55
postorder: 40 55 60 50
```

INPUT

Your INPUT goes
here!

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```
10 root = temp;
21 return root;
22 }
23 else
24 {
25     if (ele < root->data)
26         root->left = create(root->left, ele);
27     else if (ele > root->data)
28         root->right = create(root->right, ele);
29
30     return root;
31 }
32 }
33
34 void inorder(struct node *root)
35 {
36     if (root != NULL)
37     {
38         inorder(root->left);
39         printf("%d ", root->data);
40         inorder(root->right);
41     }
42 }
```

OUTPUT

```
inorder: 40 50 55 60
preorder: 50 40 60 55
postorder: 40 55 60 50
```

INPUT

Your INPUT goes
here!

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PROGRAM EDITOR C Run Save

```
40 inorder(root->right);
41 }
42 }
43
44 void preorder(struct node *root)
45 {
46     if (root != NULL)
47     {
48         printf("%d", root->data);
49         preorder(root->left);
50         preorder(root->right);
51     }
52 }
53
54 void postorder(struct node *root)
55 {
56     if (root != NULL)
57     {
58         postorder(root->left);
59         postorder(root->right);
60         printf("%d", root->data);
61     }
62 }
```

OUTPUT

```
inorder: 40 50 55 60
preorder: 50 40 60 55
postorder: 40 55 60 50
```

INPUT

Your INPUT goes here!

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PROGRAM EDITOR C Run Save

```
60 printf("%d", root->data);
61 }
62 }
63
64 int main()
65 {
66     root = create(root, 50);
67     root = create(root, 40);
68     root = create(root, 60);
69     root = create(root, 55);
70
71     printf("\ninorder:");
72     inorder(root);
73
74     printf("\npreorder:");
75     preorder(root);
76
77     printf("\npostorder:");
78     postorder(root);
79
80     return 0;

```

OUTPUT

```
inorder: 40 50 55 60
preorder: 50 40 60 55
postorder: 40 55 60 50

```

INPUT

Your INPUT goes here!